

Name: _____

In-class midterm, EC421, Spring 2026

155 points possible

1 True or false (75 points; 30 questions)

Note In this section, select the correct answer (true or false). You do not need to explain your answer.

1. (2.5 points) **[T/F]** A *population parameter* is fixed, while a *sample estimate* can change from sample to sample.
2. (2.5 points) **[T/F]** A *standard error* is the average difference between the observed outcome (y_i) and the fitted value from a regression ($\hat{y}_i$).
3. (2.5 points) **[T/F]** Ordinary least squares (OLS) requires that the regressors have variability.
4. (2.5 points) **[T/F]** If $E[u_i \mid X_i] = 0$, then the disturbance satisfies the OLS assumption of *exogeneity*.
5. (2.5 points) **[T/F]** A large p -value is strong evidence against the null hypothesis.
6. (2.5 points) **[T/F]** An LM test can jointly test several coefficients at the same time.
7. (2.5 points) **[T/F]** Adding a regressor to an OLS regression will never decrease R^2 .
8. (2.5 points) **[T/F]** Adjusted R^2 must increase whenever we add another regressor to a model.
9. (2.5 points) **[T/F]** In a multiple regression, the coefficient on x_1 describes the relationship between x_1 and y *after accounting for their relationships with the other regressors*.
10. (2.5 points) **[T/F]** The model below allows the effect of education on wages to differ by whether the individual is female:

$$\text{Wages}_i = \beta_0 + \beta_1 \text{Education}_i + \beta_2 \text{Female}_i + u_i$$

11. (2.5 points) **[T/F]** In the model below, the effect of education on wages depends on the individual's level of education:

$$\text{Wages}_i = \beta_0 + \beta_1 \text{Education}_i + \beta_2 \text{Education}_i^2 + u_i$$

12. (2.5 points) **[T/F]** In the model below, the effect of experience on wages differs between union and non-union workers:

$$\text{Wages}_i = \beta_0 + \beta_1 \text{Experience}_i + \beta_2 \text{Union}_i + \beta_3 \text{Experience}_i \times \text{Union}_i + u_i$$

13. (2.5 points) **[T/F]** In the following log-log model, a one-percent increase in x corresponds to a β_1 percent increase in y :

$$\log(y_i) = \beta_0 + \beta_1 \log(x_i) + u_i$$

14. (2.5 points) **[T/F]** Omitted-variable bias occurs whenever we omit a variable from a regression.
15. (2.5 points) **[T/F]** Classical measurement error in an *explanatory variable* biases the coefficient on that explanatory variable toward zero.
16. (2.5 points) **[T/F]** Classical measurement error in the *outcome variable* generally biases OLS slope coefficients toward zero.
17. (2.5 points) **[T/F]** *Homoskedasticity* means the individuals' disturbances have the same variance.
18. (2.5 points) **[T/F]** Heteroskedasticity causes OLS coefficient estimates to be biased—even when exogeneity holds.
19. (2.5 points) **[T/F]** Heteroskedasticity can make "classical" OLS standard errors biased.
20. (2.5 points) **[T/F]** Residuals can be used to estimate unobservable disturbances and their variances.
21. (2.5 points) **[T/F]** The Goldfeld-Quandt test compares residual variance across two subsets of the data.
22. (2.5 points) **[T/F]** The White test regresses the squared outcome, y_i^2 , on the regressors, their squares, and their interactions.

23. (2.5 points) **[T/F]** Suppose the p -value for the White test is 0.01. We should conclude there is statistically significant evidence of heteroskedasticity.
24. (2.5 points) **[T/F]** The Goldfeld-Quandt test is designed to flexibly detect multiple patterns of heteroskedasticity.
25. (2.5 points) **[T/F]** Weighted least squares (WLS) gives less weight to observations whose disturbances have larger variances.
26. (2.5 points) **[T/F]** Heteroskedasticity-robust standard errors also adjust the OLS coefficient estimates.
27. (2.5 points) **[T/F]** If disturbances are correlated within groups, then clustered standard errors can be appropriate.
28. (2.5 points) **[T/F]** Consistency describes whether an estimator is correct *on average* in a fixed sample size.
29. (2.5 points) **[T/F]** If $\text{plim } \hat{\beta}_1 \neq \beta_1$, then $\hat{\beta}_1$ is consistent for β_1 .
30. (2.5 points) **[T/F]** Omitted-variable bias means exogeneity is violated.

2 Multiple choice (20 points; 5 questions)

Note In this section, check (✓ or ×) **all** correct answers. You do not need to explain your answer.

31. (4 points) **[Multiple choice]** Choose *all* correct answers:

Which of the following are part of the standard assumptions/requirements for OLS regression?

- ☐ $E[u_i | X_i] = 0$
- ☐ There is no perfect collinearity among the regressors.
- ☐ $\text{Var}(u_i | X_i) = 0$
- ☐ $\text{Cov}(u_i, u_j | X) = 0$ for $i \neq j$

32. (4 points) **[Multiple choice]** Choose *all* correct answers:

Which of the following generally cause OLS coefficient estimates to be biased or inconsistent?

- ☐ $E[u_i | X_i] \neq 0$
- ☐ Heteroskedastic disturbances, with exogeneity still satisfied.
- ☐ An omitted variable that affects y_i and correlates with an included regressor.
- ☐ Classical measurement error in an explanatory variable.

33. (4 points) **[Multiple choice]** Choose *all* correct answers:

Suppose we estimate

$$(\text{Test Score})_i = \beta_0 + \beta_1 \text{Ratio}_i + \beta_2 \text{Income}_i + u_i.$$

Which items belong in the regression for the White test (as dependent or independent variables)?

- ☐ e_i^2
- ☐ Ratio_i^2
- ☐ $\text{Ratio}_i \times \text{Income}_i$
- ☐ $(\text{Test Score})_i^2$

34. (4 points) **[Multiple choice]** Choose *all* correct answers:

In the presence of heteroskedasticity, which of the following are reasonable responses?

- ☐ Check the model specification.
- ☐ Use heteroskedasticity-robust standard errors for inference.
- ☐ Use WLS if you know the form of the heteroskedasticity.
- ☐ Use log transformations.

35. (4 points) **[Multiple choice]** Choose *all* correct answers:

Consider the model

$$\log(\text{Wage}_i) = \beta_0 + \beta_1 \text{College}_i + \beta_2 \text{Experience}_i + \beta_3 \text{College}_i \times \text{Experience}_i + u_i,$$

where College_i is a binary variable and Experience_i is the number of years of experience. Which descriptions are correct?

- ☐ β_2 tells us how an additional year of experience translates into a percent increase for the wages of non-college workers.
- ☐ β_3 describes how the effect of experience differs for college workers relative to non-college workers.
- ☐ β_1 gives the effect of college graduation.
- ☐ β_3 describes how an additional year of experience translates into a percent increase for the wages of college workers.

3 Short answer (60 points; 12 questions)

Note In this section, briefly answer the questions/prompts in 1–3 short (and complete) sentences.

We will deduct points for excessively long answers.

36. (5 points) Define the concept of a *standard error*.

37. (5 points) Explain the difference between a *disturbance* and a *residual*.

38. (5 points) Explain the meaning of *correlated disturbances* and discuss their implications for OLS inference.

39. (5 points) We want to estimate the effect of class size on test scores using

$$\text{Score}_i = \beta_0 + \beta_1(\text{Class Size})_i + \beta_2(\text{Family Income})_i + u_i,$$

but we do not observe family income. Suppose family income has a positive effect on test scores and schools with larger classes tend to have lower family incomes. Use the omitted-variable-bias formula below to sign the bias from omitting family income. Is OLS more likely to overestimate or underestimate the class-size coefficient when we omit family income? Explain your answer.

$$\text{plim } \hat{\beta}_1 = \beta_1 + \beta_2 \frac{\text{Cov}(x_1, x_2)}{\text{Var}(x_1)}$$

40. (5 points) Suppose we estimate

$$\text{Score}_i = \beta_0 + \beta_1 \text{Ratio}_i + \beta_2 \text{Income}_i + u_i$$

and obtain $\hat{\beta}_1 = -3.5$ with standard error 0.5. Explain whether the estimate is likely to be statistically significant at the 5-percent level.

41. (5 points) Can increasing the sample size solve the omitted-variable-bias problem above? Explain.

42. (5 points) What are the main consequences of heteroskedasticity for OLS (1) coefficient estimates, (2) efficiency, and (3) inference?

43. (5 points) Compare the Goldfeld-Quandt test and the White test for heteroskedasticity.

44. (5 points) Consider the model

$$\text{Health}_i = \beta_0 + \beta_1 \text{Exercise}_i + \beta_2 \text{Senior}_i + \beta_3 \text{Exercise}_i \times \text{Senior}_i + u_i,$$

where Health_i is a health index, Exercise_i is hours of exercise per week, and Senior_i equals 1 for senior citizens and 0 otherwise. Suppose $\beta_0 = 60$, $\beta_1 = 3$, $\beta_2 = -12$, and $\beta_3 = -1$. Interpret β_1 and β_3 *using the equation and the provided numbers*.

45. (5 points) Suppose you work for a restaurant company with franchise locations across the country. You have data on the current sales revenue, prices, and advertising expenditures for each franchise. Your boss wants you to estimate the effects of pricing and advertising on sales using the regression model below.

$$\text{Sales}_i = \beta_0 + \beta_1 \text{Price}_i + \beta_2 \text{Advertising}_i + u_i$$

What econometric concerns should you raise about this regression model?

46. (5 points) Draw a plot that shows disturbances that are exogenous and heteroskedastic.

47. (5 points) Draw a plot that shows disturbances that are endogenous and homoskedastic.